

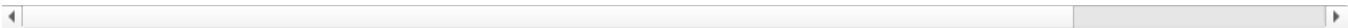
```
In [10]: from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score
from sklearn.model_selection import train_test_split
```

```
In [11]: import pandas as pd
data=pd.read_csv("Housing.csv")
data
```

Out[11]:

	price	area	bedrooms	bathrooms	stories	mainroad	guestroom	basement	hotwaterheating	airconditioning	parking	pr
0	13300000	7420	4	2	3	yes	no	no	no	yes	2	
1	12250000	8960	4	4	4	yes	no	no	no	yes	3	
2	12250000	9960	3	2	2	yes	no	yes	no	no	2	
3	12215000	7500	4	2	2	yes	no	yes	no	yes	3	
4	11410000	7420	4	1	2	yes	yes	yes	no	yes	2	
...	...	...	...	...	...	...	...	...	...	...	...	...
540	1820000	3000	2	1	1	yes	no	yes	no	no	2	
541	1767150	2400	3	1	1	no	no	no	no	no	0	
542	1750000	3620	2	1	1	yes	no	no	no	no	0	
543	1750000	2910	3	1	1	no	no	no	no	no	0	
544	1750000	3850	3	1	2	yes	no	no	no	no	0	

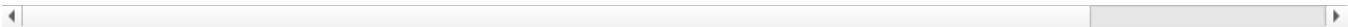
545 rows × 13 columns



```
In [12]: data.head(5) #first 5 rows will be printed.
```

Out[12]:

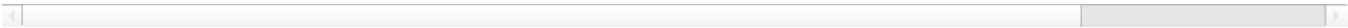
	price	area	bedrooms	bathrooms	stories	mainroad	guestroom	basement	hotwaterheating	airconditioning	parking	pre
0	13300000	7420	4	2	3	yes	no	no	no	yes	2	
1	12250000	8960	4	4	4	yes	no	no	no	yes	3	
2	12250000	9960	3	2	2	yes	no	yes	no	no	2	
3	12215000	7500	4	2	2	yes	no	yes	no	yes	3	
4	11410000	7420	4	1	2	yes	yes	yes	no	yes	2	



```
In [13]: data.head(10)
```

Out[13]:

	price	area	bedrooms	bathrooms	stories	mainroad	guestroom	basement	hotwaterheating	airconditioning	parking	pr
0	13300000	7420	4	2	3	yes	no	no	no	yes	2	
1	12250000	8960	4	4	4	yes	no	no	no	yes	3	
2	12250000	9960	3	2	2	yes	no	yes	no	no	2	
3	12215000	7500	4	2	2	yes	no	yes	no	yes	3	
4	11410000	7420	4	1	2	yes	yes	yes	no	yes	2	
5	10850000	7500	3	3	1	yes	no	yes	no	yes	2	
6	10150000	8580	4	3	4	yes	no	no	no	yes	2	
7	10150000	16200	5	3	2	yes	no	no	no	no	0	
8	9870000	8100	4	1	2	yes	yes	yes	no	yes	2	
9	9800000	5750	3	2	4	yes	yes	no	no	yes	1	



```
In [4]: data.shape #tells us the number of rows and columns present in the csv file.
```

Out[4]: (545, 13)

```
In [5]: data.info() #this returns not null values,column,datatype,and information about the data.
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 545 entries, 0 to 544
Data columns (total 13 columns):
#   Column                Non-Null Count  Dtype
---  -
0   price                 545 non-null   int64
1   area                 545 non-null   int64
2   bedrooms             545 non-null   int64
3   bathrooms            545 non-null   int64
4   stories              545 non-null   int64
5   mainroad             545 non-null   object
6   guestroom           545 non-null   object
7   basement             545 non-null   object
8   hotwaterheating      545 non-null   object
9   airconditioning      545 non-null   object
10  parking              545 non-null   int64
11  prefarea             545 non-null   object
12  furnishingstatus     545 non-null   object
dtypes: int64(6), object(7)
memory usage: 55.5+ KB
```

```
In [16]: from sklearn.preprocessing import LabelEncoder, MinMaxScaler #this command will convert object datatype into int
le=LabelEncoder() #it converts the categorical entries into numerical entries.
data["mainroad"]=le.fit_transform(data["mainroad"])
data
#change raw feature vectors into a representation that is more suitable for the downstream estimators-sklearn.p
```

```
Out[16]:
```

	price	area	bedrooms	bathrooms	stories	mainroad	guestroom	basement	hotwaterheating	airconditioning	parking	p
0	13300000	7420	4	2	3	1	no	no	no	yes	2	
1	12250000	8960	4	4	4	1	no	no	no	yes	3	
2	12250000	9960	3	2	2	1	no	yes	no	no	2	
3	12215000	7500	4	2	2	1	no	yes	no	yes	3	
4	11410000	7420	4	1	2	1	yes	yes	no	yes	2	
...	...	...	...	...	...	...	...	...	...	...	...	
540	1820000	3000	2	1	1	1	no	yes	no	no	2	
541	1767150	2400	3	1	1	0	no	no	no	no	0	
542	1750000	3620	2	1	1	1	no	no	no	no	0	
543	1750000	2910	3	1	1	0	no	no	no	no	0	
544	1750000	3850	3	1	2	1	no	no	no	no	0	

545 rows × 13 columns

```
In [17]: from sklearn.preprocessing import LabelEncoder, MinMaxScaler
data["guestroom"]=le.fit_transform(data["guestroom"])
data.head(5)
```

```
Out[17]:
```

	price	area	bedrooms	bathrooms	stories	mainroad	guestroom	basement	hotwaterheating	airconditioning	parking	pre
0	13300000	7420	4	2	3	1	0	no	no	yes	2	
1	12250000	8960	4	4	4	1	0	no	no	yes	3	
2	12250000	9960	3	2	2	1	0	yes	no	no	2	
3	12215000	7500	4	2	2	1	0	yes	no	yes	3	
4	11410000	7420	4	1	2	1	1	yes	no	yes	2	

```
In [18]: from sklearn.preprocessing import LabelEncoder, MinMaxScaler
data["basement"]=le.fit_transform(data["basement"])
data.head(5)
```

```
Out[18]:
```

	price	area	bedrooms	bathrooms	stories	mainroad	guestroom	basement	hotwaterheating	airconditioning	parking	pre
0	13300000	7420	4	2	3	1	0	0	no	yes	2	
1	12250000	8960	4	4	4	1	0	0	no	yes	3	
2	12250000	9960	3	2	2	1	0	1	no	no	2	
3	12215000	7500	4	2	2	1	0	1	no	yes	3	
4	11410000	7420	4	1	2	1	1	1	no	yes	2	

```
In [19]: from sklearn.preprocessing import LabelEncoder,MinMaxScaler
data["hotwaterheating"]=le.fit_transform(data["hotwaterheating"])
data.head(5)
```

```
Out[19]:
```

	price	area	bedrooms	bathrooms	stories	mainroad	guestroom	basement	hotwaterheating	airconditioning	parking	pre
0	13300000	7420	4	2	3	1	0	0	0	yes	2	
1	12250000	8960	4	4	4	1	0	0	0	yes	3	
2	12250000	9960	3	2	2	1	0	1	0	no	2	
3	12215000	7500	4	2	2	1	0	1	0	yes	3	
4	11410000	7420	4	1	2	1	1	1	0	yes	2	

```
In [27]: from sklearn.preprocessing import LabelEncoder,MinMaxScaler
data["prefarea"]=le.fit_transform(data["prefarea"])
data.head(5)
```

```
Out[27]:
```

	price	area	bedrooms	bathrooms	stories	mainroad	guestroom	basement	hotwaterheating	airconditioning	parking	pre
0	13300000	7420	4	2	3	1	0	0	0	1	2	
1	12250000	8960	4	4	4	1	0	0	0	1	3	
2	12250000	9960	3	2	2	1	0	1	0	0	2	
3	12215000	7500	4	2	2	1	0	1	0	1	3	
4	11410000	7420	4	1	2	1	1	1	0	1	2	

```
In [28]: from sklearn.preprocessing import LabelEncoder,MinMaxScaler
data["furnishingstatus"]=le.fit_transform(data["furnishingstatus"])
data.head(5)
```

```
Out[28]:
```

	price	area	bedrooms	bathrooms	stories	mainroad	guestroom	basement	hotwaterheating	airconditioning	parking	pre
0	13300000	7420	4	2	3	1	0	0	0	1	2	
1	12250000	8960	4	4	4	1	0	0	0	1	3	
2	12250000	9960	3	2	2	1	0	1	0	0	2	
3	12215000	7500	4	2	2	1	0	1	0	1	3	
4	11410000	7420	4	1	2	1	1	1	0	1	2	

```
In [29]: x=data.drop(columns=["price"])
y=data["price"]
y=y.values.reshape(-1,1)
```

```
In [31]: scaler=MinMaxScaler()
x=scaler.fit_transform(x)
y=scaler.fit_transform(y)
```

```
In [33]: lr=LinearRegression()
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.2)
lr.fit(x_train,y_train)
y_predict=lr.predict(x_test)
```

```
In [37]: mae=mean_absolute_error(y_test,y_predict)
mse=mean_squared_error(y_test,y_predict)
r2=r2_score(y_test,y_predict)
print(mae,mse,r2)
```

0.06995281320799962 0.007960782075320859 0.6594122430015953